Experiment-6
I-V Characteristics of MOSFET

Name: S.Vathsalya
Roll.No:se25mvls020
Course :M.Tech VLSI &ES

Aim: Design an n-channel Metal Oxide Semiconductor Field Effect Transistor (nMOSFET) and simulate the transfer (I_D–V_{GS}) and output (I_D–V_{DS}) characteristics using Synopsys TCAD Sentaurus.

Objectives:

1 Study the I_D–V_{GS} characteristics of n-MOSFET for different values of V_{DS}.
2 Study the I_D–V_{DS} characteristics of n-MOSFET for different values of V_{GS}.
3 Study the effect of device dimensions on the I_D–V_{GS} characteristics.
4 Study the effect of different materials on the I_D–V_{GS} characteristics.
5 Optimized the device structure for higher ION/IOFF ratio.

Software Tool:
Synopsys TCAD Sentaurus.

Theory: The metal–oxide–semiconductor field-effect transistor (MOSFET) is a transistor used for amplifying or switching electronic signals. In MOSFETs, a voltage on the oxide-insulated gate electrode can induce a conducting channel between the two other contacts called source and drain. The channel canbe of n-type or p-type, and is accordingly called an nMOSFET or a pMOSFET.

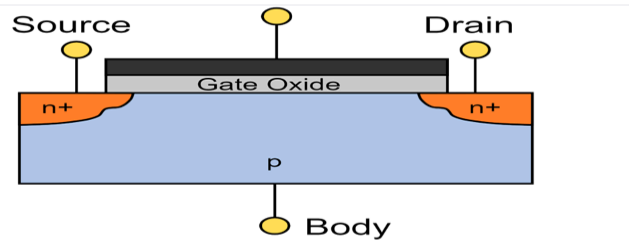


Fig :NMOSFET Structure

Procedure:
1.Create a new folder. Open Terminal in the new folder then enter “sde &” command in the terminal.
2.Go to View a GUI Mode a 2D Mode.
3.Go to Draw then deselect the “Auto Region Naming “and select the “Exact Coordinates”.
4.Now go to materials and select “silicon” (silicon is the default material) then click on “Create rectangular Sheet Region” on the toolbar.
5.Place the rectangle on Canvas with (0,0.045) and (0,0.180) coordinates and press ok.
6.Now enter Region Name as “substrate”.
7.Dope the substrate with Boron Active concentration of 1e18. For doping the silicon region, Go to Device a Constant Profile Placement.
8.Deposit oxide layer source on the substrate. Enter the region name as source.with coordinates(0.00,0.025) and (0.0,0.06).
9.Dope the source with Phosphorous Active concentration of 1e20. For doping the silicon

region, Go to Device a Constant Profile Placement.

10.Deposit oxide layer drain on the substrate. Enter the region name as drain.with coordinates(0.00,0.025) and (0.180,0.120).

11.Dope the Drain with Phosphorous Active concentration of 1e19. For doping the silicon region, Go to Device a Constant Profile Placement.

12..Deposit oxide layer sio2 on the substrate. Enter the region name as oxide.with coordinates(0.00,0.060) and (-0.045,0.120).

13.Deposit metal layer on oxide layer. Choose GatePolySi as metal. Enter the region name as metal with coordinates (-0.0045,-0.009) and (0.060,0.120).

14.Give the contacts to gate source drain body. For creating Contact, Go to Contacts a Contact Sets.Enter the contact name as gatec ,bodyc,drainc,sourcec with different edge colour.

15.To Set Contact first change the Select Body to Select Edge on the Tool Bar.

16.Now click on Arrow () placed at the left panel and go to Contacts a Set Contact then place the electrical contact on the left and right edges of silicon.

17.Go to Mesh a Define Rel/Eval Window and select rectangle à draw rectangle on canvas covering the entire design.

18.Got Mesh à Refinement Placement enter the minimum & maximum element size. Click on create Refinement.

19.Go to Mesh à click on Build Mesh. Then the device structure will be open in the svisual.

20.Got Mesh à Refinement Placement enter the minimum & maximum element size.minimum size is 1e-06 and 0.005 is the maximum element size and Click on create Refinement.

21.Go to Mesh click on Build Mesh. Then the device structure will be open in svisual.

22.To study the device characteristics, edit the nmos “cmd file” (.cmd file). msh.tdr file name should be included in cmd file Grid section.

23. Edit nmos. cmd file for different sweep voltages from 0V to 2V.

24.Enter the “sdevice cmdfilename” (Ex: sdevice TEST_nmos.cmd) in working directory terminal.

25.To visualize the device characteristics, enter svisual command in terminal.

26.Go to file in svisual and open plot file.

27.For plotting I_D–V_{GS} characteristics select gatec outer voltage to x-axis and drainc total current to left y-axis.

RESULT:

Fig 1a: I_D–V_{GS} characteristics of n-MOSFET for different values of V_{DS}

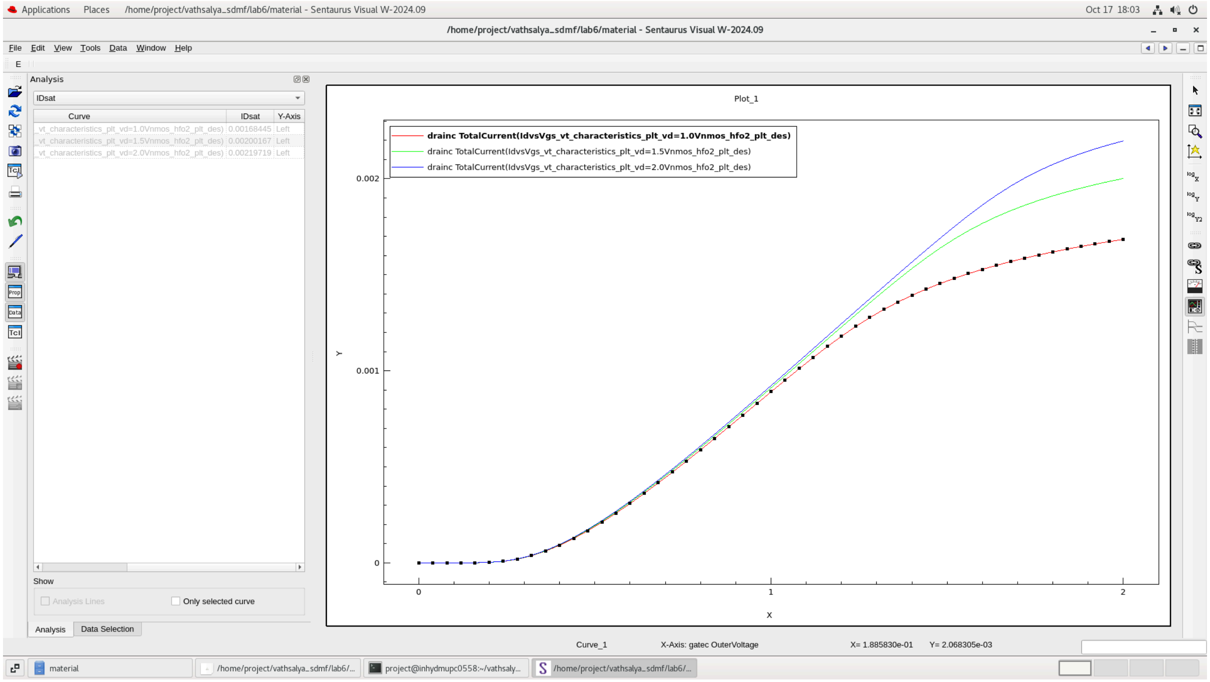


Fig 1b: I_D–V_{GS} characteristics of n-MOSFET for different values of V_{DS}

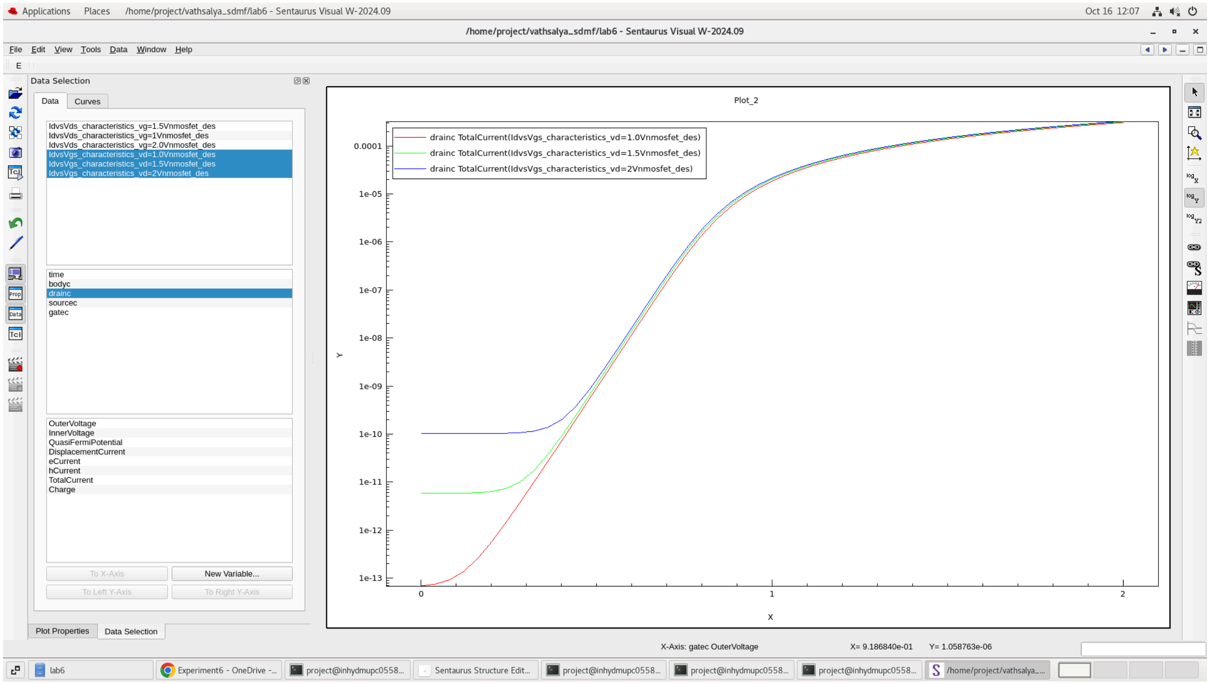


Fig 2: I_D–V_{DS} characteristics of n-MOSFET for different values of V_{GS}

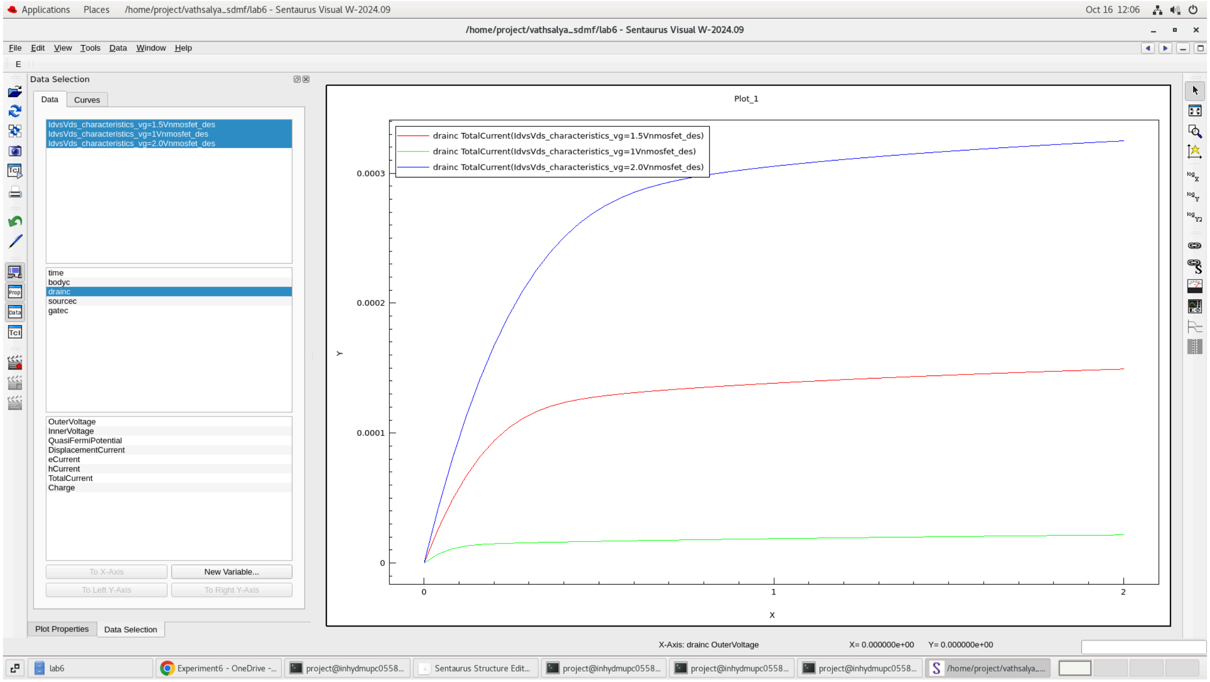


Fig 3a:device dimensions on the I_D–V_{GS} characteristics

THE DIMENSIONS ARE REDUCED FROM 0.120µm to 0.06µm

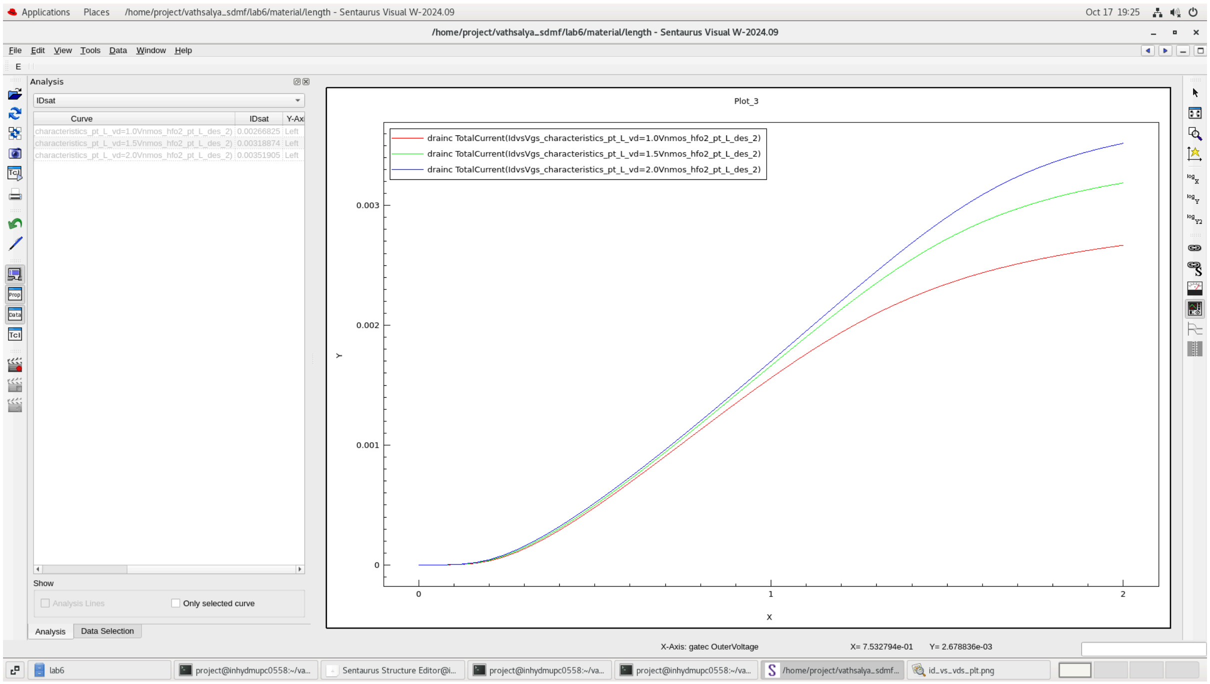


Fig 3b: Mesh diagram of reduced channel length

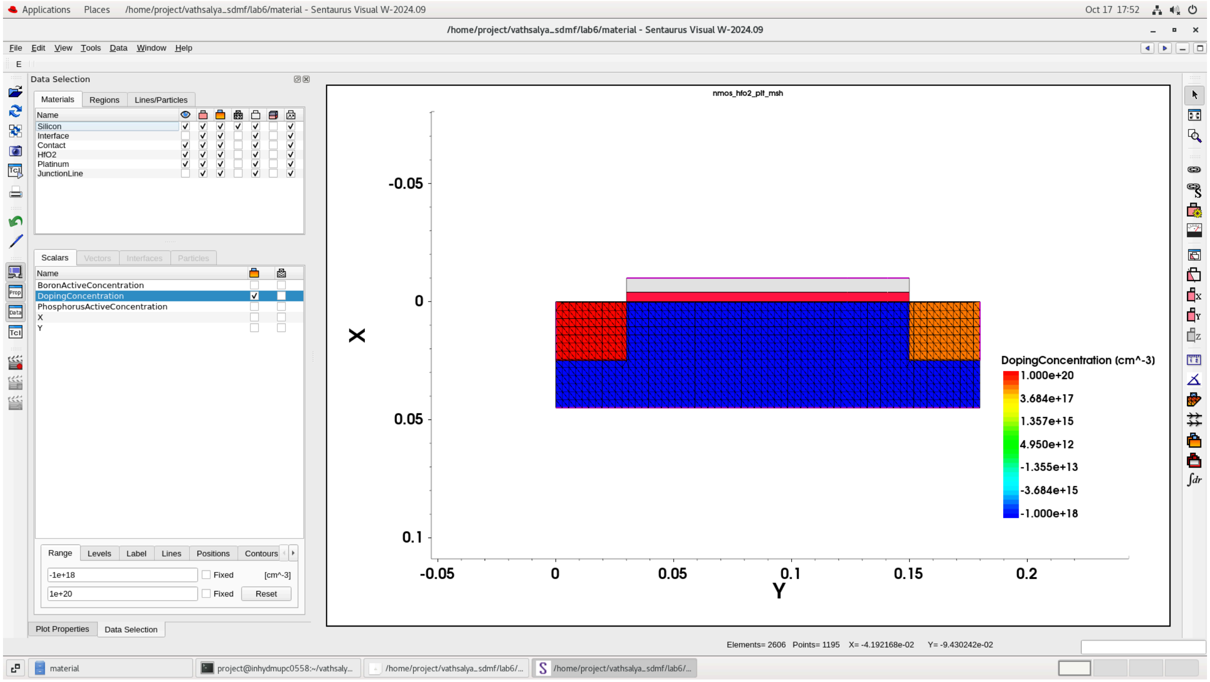


Fig 4a:Effect of nickel on the I_D–V_{GS} characteristics

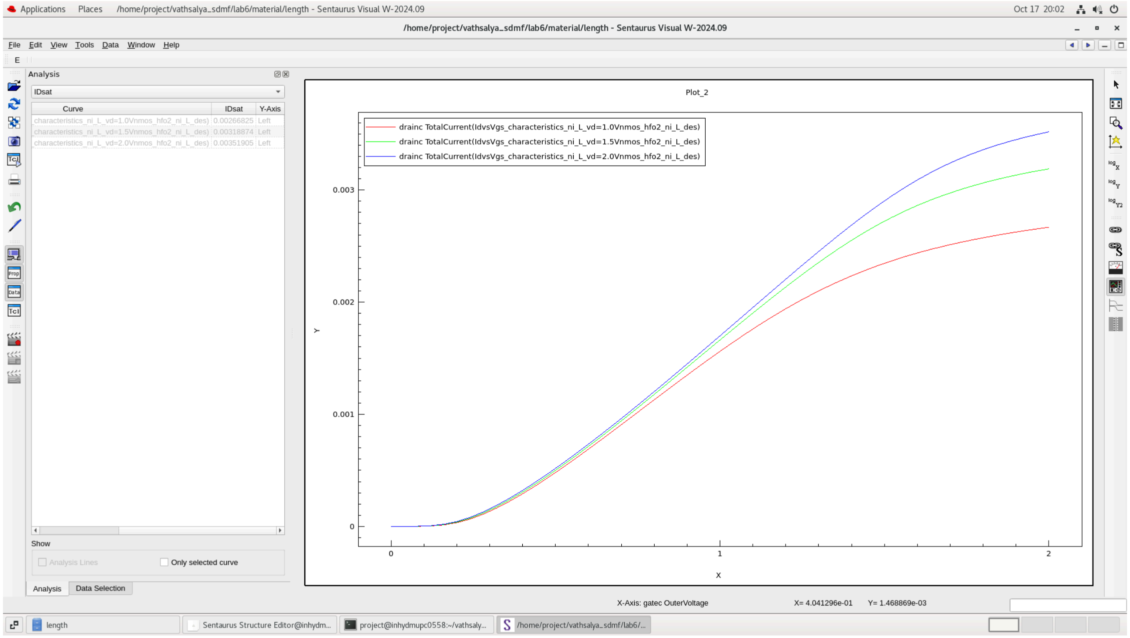


Fig 4b:Mesh Diagram

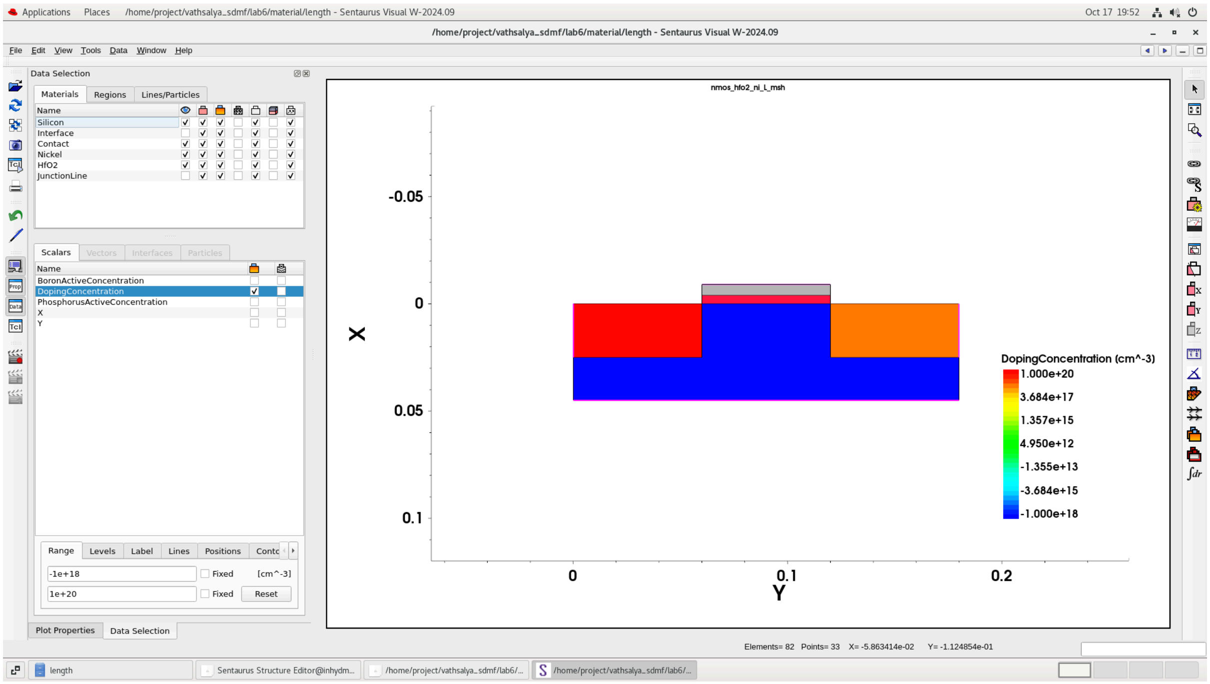


Fig 5a: Effect of aluminium on the I_D–V_{GS} characteristics

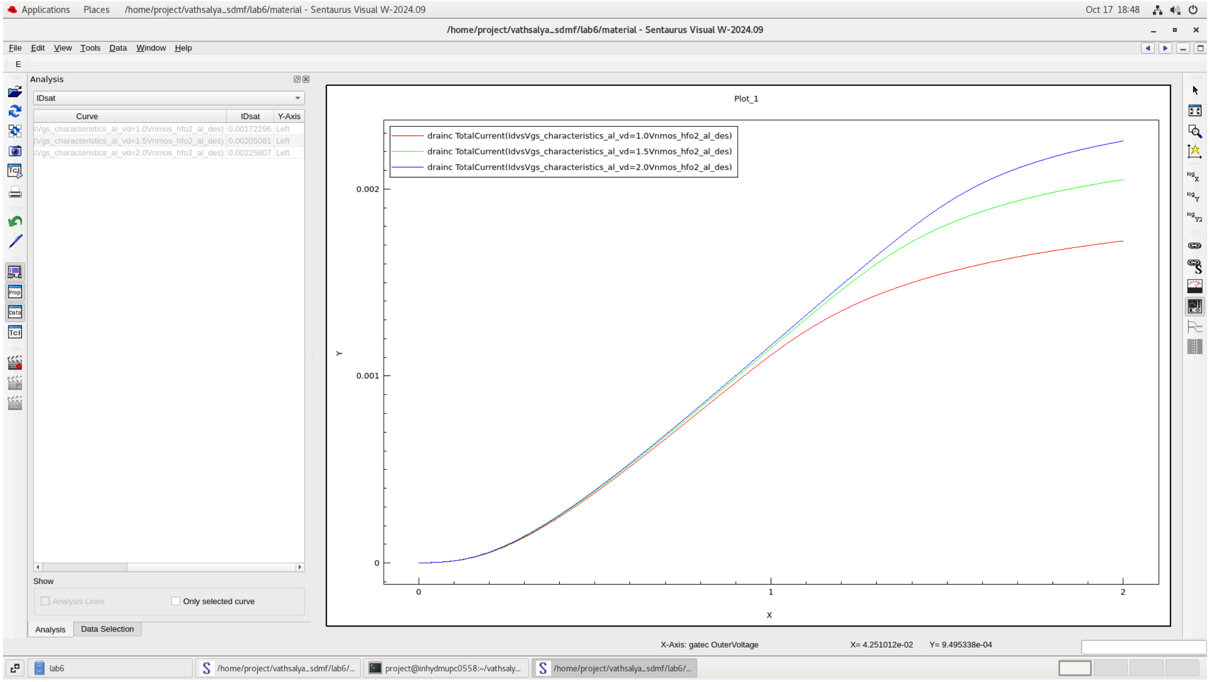


Fig 5b:Mesh Diagram of Aluminium as gate

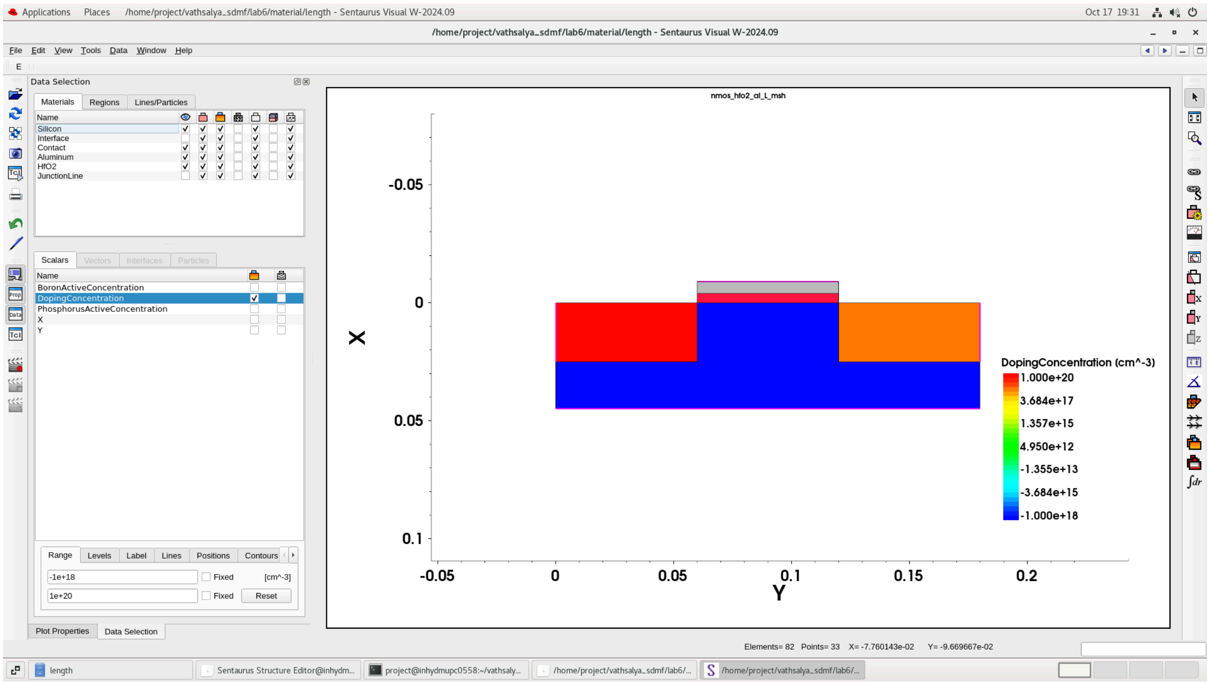


Fig 6a:Effect of Platinum on the I_D–V_{GS} characteristics

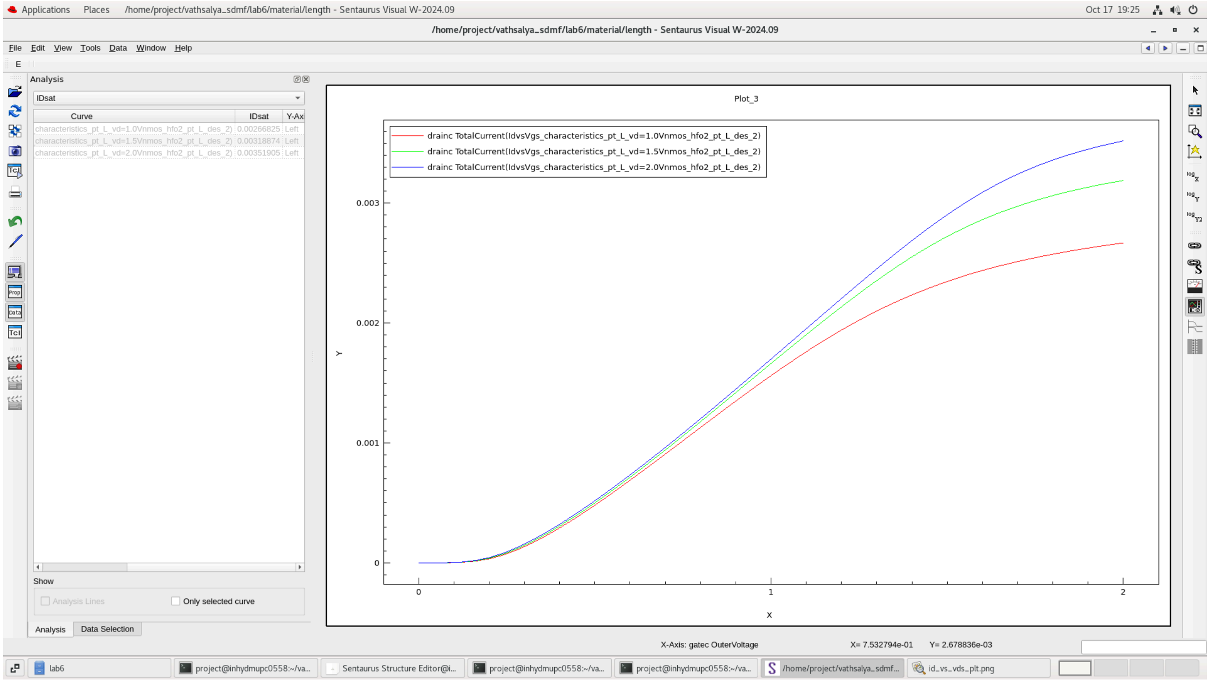
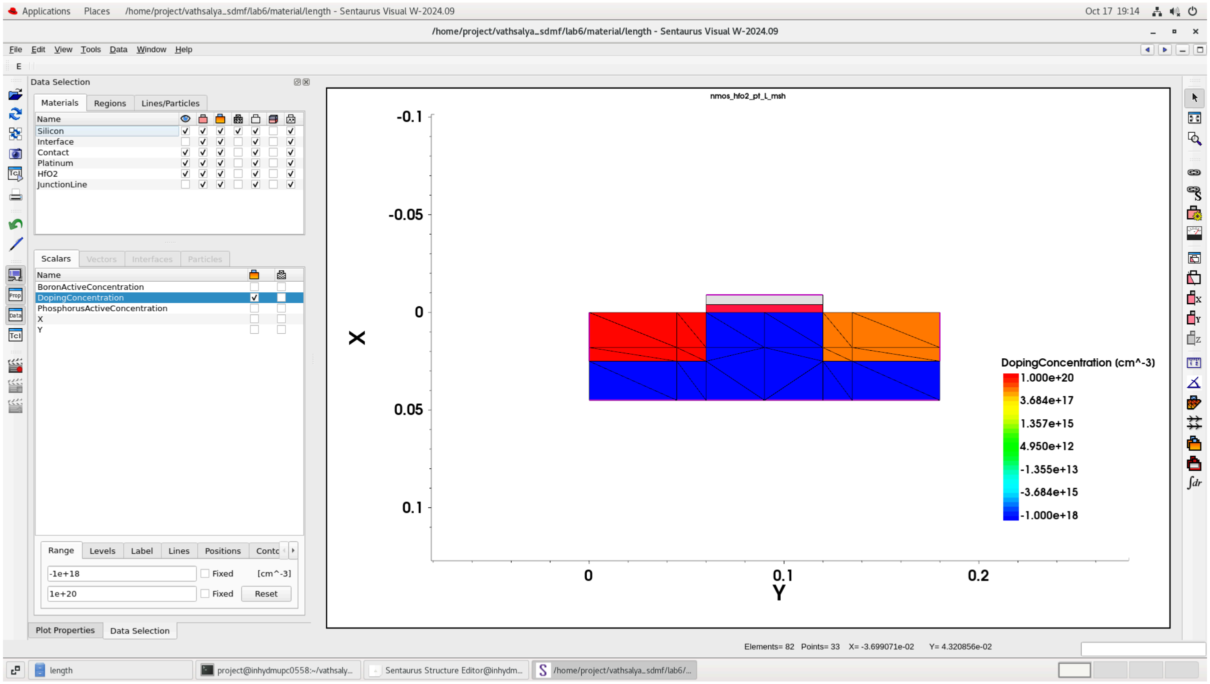


Fig 6b:Mesh Diagram of Platinum



OBSERVATIONS:

1.As VGS increases, the drain current ID increases for all values of VDS.

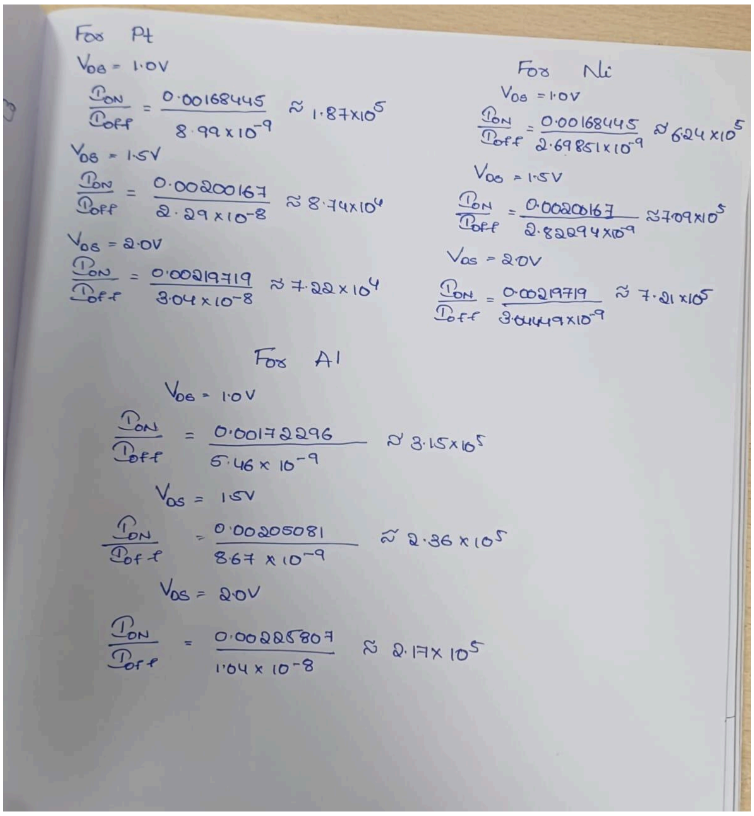
For higher VDS, the ID–VGS curve rises to higher current levels because the channel is more strongly inverted.

Below threshold voltage, current is almost zero.

2.For small VDS, ID increases almost linearly with VDS → **ohmic/linear region**.

After a certain point (when VDS ≥ VGS − VTH), ID becomes almost constant → **saturation region**.
Upon increasing the value of VGS the VDS value becomes negative and the electrons in the channel get repelled by the negative voltage of VDS.Then the channel gets pinched off.

3.When the channel length is reduced from 0.120µm to 0.06µm the current is inversely proportional to the resistance where resistance is directly proportional to the length so when length is reduced resistance is reduced so the current is increased.
4.1.Platinum (5.6eV)
->Very high threshold voltage (VTH)
->Channel forms later
->ID starts rising at higher VGS
2.Nickel (5.2eV)
->Medium VTH
->channel forms
->Moderate ION
3.Aluminium (4.1eV)
->Lowest threshold voltage
->Channel forms early
->ID rises early even at small VGS
->Gives highest ION
5.Optimised ion/ioff



For Pt

$V_{DS} = 1.0V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00168445}{8.99 \times 10^{-9}} \approx 1.87 \times 10^5$$

$V_{DS} = 1.5V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00200167}{2.29 \times 10^{-8}} \approx 8.74 \times 10^4$$

$V_{DS} = 2.0V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00219719}{3.04 \times 10^{-8}} \approx 7.22 \times 10^4$$

For Ni

$V_{DS} = 1.0V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00168445}{2.69851 \times 10^{-9}} \approx 6.24 \times 10^5$$

$V_{DS} = 1.5V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00200167}{2.82294 \times 10^{-9}} \approx 7.09 \times 10^5$$

$V_{DS} = 2.0V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00219719}{3.04449 \times 10^{-9}} \approx 7.21 \times 10^5$$

For Al

$V_{DS} = 1.0V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00172296}{5.46 \times 10^{-9}} \approx 3.15 \times 10^5$$

$V_{DS} = 1.5V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00205081}{8.67 \times 10^{-9}} \approx 2.36 \times 10^5$$

$V_{DS} = 2.0V$

$$\frac{I_{ON}}{I_{off}} = \frac{0.00225807}{1.04 \times 10^{-8}} \approx 2.17 \times 10^5$$

The value of work function is more for platinum is higher than the rest of the metals so the threshold value is higher then Ion values decreases.

Conclusion:The transfer ((ID–VGS) and output (ID–VDS) characteristics of an n- channel Metal Semiconductor Field Effect Transistor (MOSFET) are studied using Synopsys TCAD Sentaurus software.